

Ex.No. 14: 5G Throughput vs SNR with Modulation Labels**Date:3-8-26**

AIM: To simulate the performance of a 5G wireless communication system using MATLAB by analyzing the relationship between Throughput (Mbps), Signal-to-Noise Ratio (SNR) (dB), and different Modulation Schemes (QPSK, 16-QAM, 64-QAM, and 256-QAM).

Objective:

1. To evaluate the Throughput (Mbps) of a 5G communication system under different Signal-to-Noise Ratio (SNR) conditions.
2. To analyze the effect of Signal-to-Noise Ratio (SNR) (dB) on the performance and reliability of the 5G communication system.
3. To compare the performance of different Modulation Schemes (QPSK, 16-QAM, 64-QAM, and 256-QAM) and determine their suitability for varying SNR conditions.

Procedure

1. Define the simulation parameters such as SNR range, bandwidth, modulation schemes (QPSK, 16-QAM, 64-QAM, and 256-QAM), and transmission settings.
2. Generate random binary data and modulate it using the selected modulation schemes.
3. Simulate the transmission of the modulated signals through an Additive White Gaussian Noise (AWGN) channel with different SNR values.
4. Calculate the Throughput (Mbps) achieved for each modulation scheme at different SNR levels.
5. Analyze the impact of Signal-to-Noise Ratio (SNR) on communication performance and throughput.
6. Compare the performance of QPSK, 16-QAM, 64-QAM, and 256-QAM under identical channel conditions.
7. Plot MATLAB graphs for Throughput vs SNR and SNR vs Modulation Scheme to visualize the system performance.
8. Interpret the results to identify the most suitable modulation scheme for different SNR conditions in a 5G communication system.

Objective 1

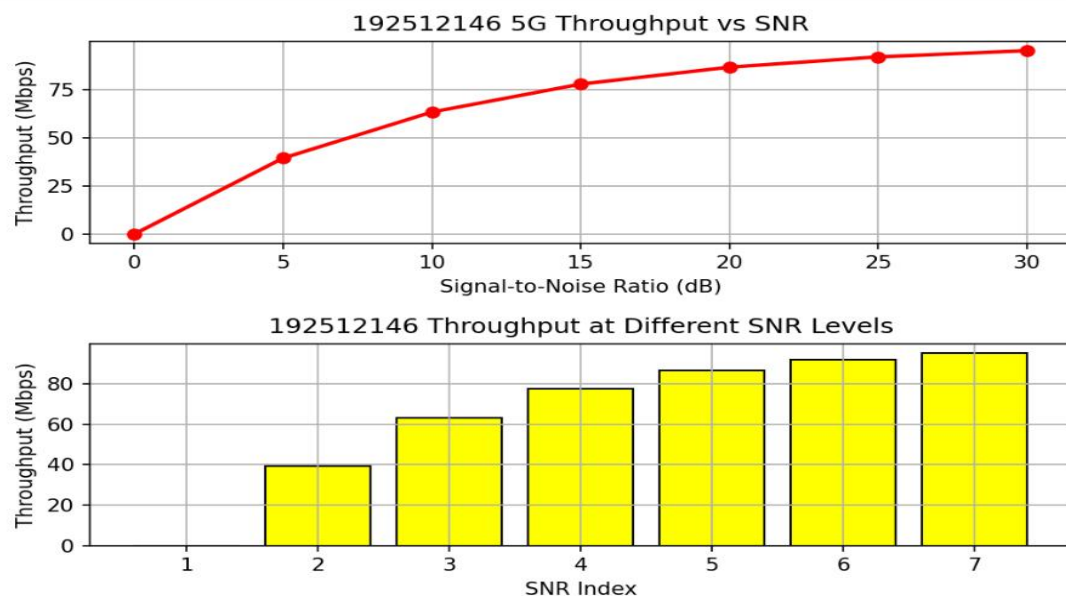
Matlab Code and Output:

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Maximum Throughput (Mbps)
Max_TP = 100;
% Throughput calculation

Throughput = Max_TP*(1-exp(-SNR/10));
disp('SNR(dB) Throughput(Mbps)')
disp([SNR' Throughput'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(SNR,Throughput,'-o','LineWidth',2)
title('5G Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on
% ----- Graph 2 -----
subplot(2,1,2)
bar(Throughput)
title('Throughput at Different SNR Levels')
xlabel('SNR Index')
```

```
ylabel('Throughput (Mbps)')
grid on
```

```
1  clc;
2  clear;
3  close all;
4  % Vehicle Speed (km/h)
5  speed = 20:20:120;
6  % Simulated End-to-End Latency (ms)
7  latency = [18 16 14 12 10 9];
8  disp('Vehicle Speed (km/h)   End-to-End Latency (ms)')
9  disp([speed' latency'])
10
11 figure
12 % ----- Graph 1 -----
13 subplot(2,1,1)
14 plot(speed,latency,'-o','LineWidth',2,'Color','b')
15 title('192512146 Vehicle Speed vs End-to-End Latency')
16 xlabel('Vehicle Speed (km/h)')
17 ylabel('Latency (ms)')
18 grid on
19 % ----- Graph 2 -----
20 subplot(2,1,2)
21 plot(1:length(latency),latency,'-s','LineWidth',2,'Color','g')
22 title('192512146 Latency Variation Across Vehicles')
23 xlabel('Vehicle Index')
24 ylabel('End-to-End Latency (ms)')
25 grid on
```



The throughput of the 5G system increases as the SNR increases due to improved signal quality.

Higher SNR enables efficient data transmission, resulting in greater throughput and better communication performance

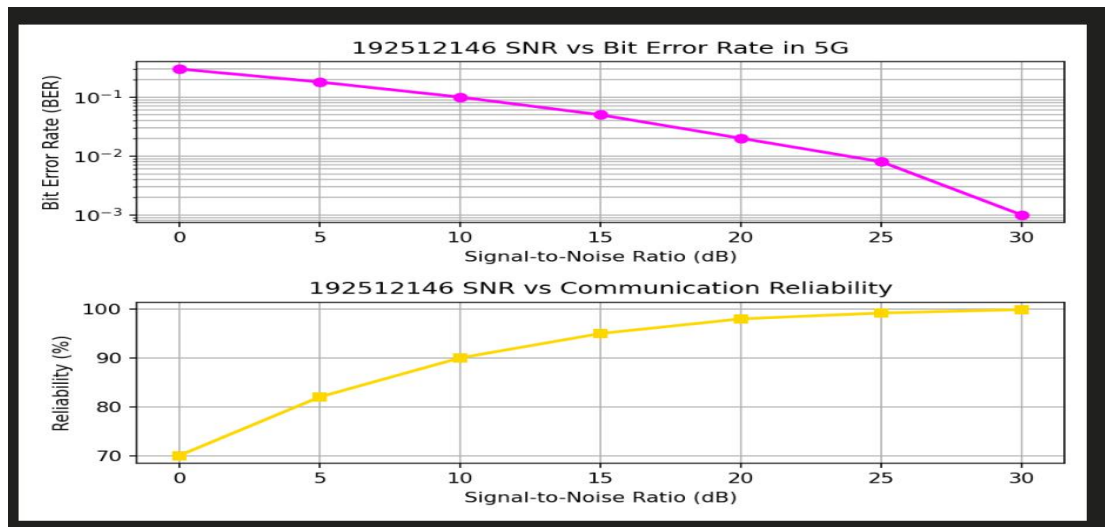
Objective 2 Matlab Code and Output:

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Bit Error Rate (BER)
BER = [0.30 0.18 0.10 0.05 0.02 0.008 0.001];
```

```

% Reliability (%)
Reliability = (1-BER)*100;
disp('SNR(dB) BER Reliability(%)')
disp([SNR' BER' Reliability'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
semilogy(SNR,BER,'-o','LineWidth',2)
title('SNR vs Bit Error Rate in 5G')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Bit Error Rate (BER)')
grid on
% ----- Graph 2 -----
subplot(2,1,2)
plot(SNR,Reliability,'-s','LineWidth',2)
title('SNR vs Communication Reliability')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Reliability (%)')
grid on
1 clc;
2 clear;
3 close all;
4 % Vehicle Speed (km/h)
5 speed = 20:20:120;
6 % Simulated Packet Delivery Ratio (%)
7 PDR = [99 98.5 97 95 93 90];
8 disp([speed' PDR'])
9
10 figure
11 % ----- Graph 1 -----
12 subplot(2,1,1)
13 plot(speed,PDR,'-o','LineWidth',2,'Color',"magenta")
14 title('192512146 Vehicle Speed vs Packet Delivery Ratio')
15 xlabel('Vehicle Speed (km/h)')
16 ylabel('Packet Delivery Ratio (%)')
17 grid on
18 % ----- Graph 2 -----
19 subplot(2,1,2)
20 plot(1:length(PDR),PDR,'-s','LineWidth',2,'Color',"r")
21 title('192512146 Packet Delivery Ratio Across Vehicles')
22 xlabel('Vehicle Index')
23 ylabel('Packet Delivery Ratio (%)')
24 grid on

```



As the Signal-to-Noise Ratio (SNR) increases, the Bit Error Rate (BER) decreases, resulting in more reliable communication.

Higher SNR improves the quality of the received signal, thereby enhancing the overall performance of the 5G communication system.

Objective 3 Matlab Code and Output:

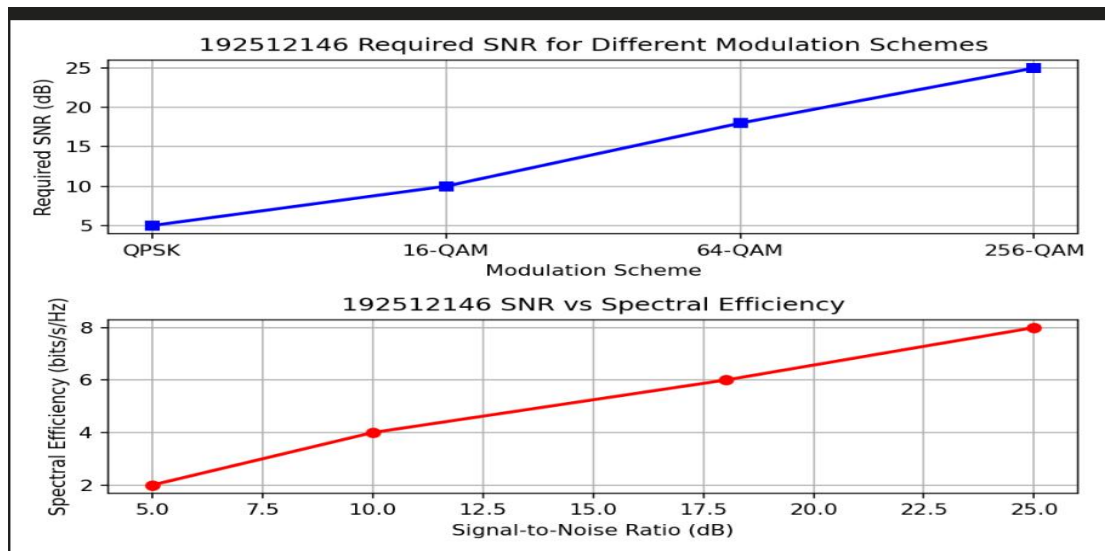
```
clc;
clear;
close all;
% Modulation schemes
Modulation = {'QPSK','16-QAM','64-QAM','256-QAM'};
% Required SNR (dB)
SNR = [5 10 18 25];
% Spectral Efficiency (bits/s/Hz)
Efficiency = [2 4 6 8];
disp('Required SNR (dB) and Spectral Efficiency')
disp(table(Modulation',SNR',Efficiency', ...
'VariableNames',{'Modulation','SNR_dB','Efficiency'}))
figure
% ----- Graph 1 -----
subplot(2,1,2)

plot(SNR,Efficiency,'-o','LineWidth',2)
title('SNR vs Spectral Efficiency')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Spectral Efficiency (bits/s/Hz)')
grid on
% ----- Graph 2 -----
subplot(2,1,1)
plot(1:4,SNR,'-s','LineWidth',2)
xticks(1:4)
xticklabels(Modulation)
title('Required SNR for Different Modulation Schemes')
xlabel('Modulation Scheme')
ylabel('Required SNR (dB)')
grid on
```

```

1  clc;
2  clear;
3  close all;
4  % Vehicle IDs
5  vehicle = 1:6;
6  % Vehicle Speed (km/h)
7  speed = [30 45 60 75 90 105];
8  % Communication Range (m)
9  range = [280 260 240 220 200 180];
10 disp(' Vehicle Speed(km/h)  Communication Range(m)')
11 disp([vehicle' speed' range'])
12 figure
13 % ----- Graph 1 -----
14 subplot(2,1,1)
15 plot(vehicle,speed,'-o','LineWidth',2,'Color',"r")
16 title('192512146 Vehicle Speed of Different Vehicles')
17 xlabel('Vehicle Index')
18 ylabel('Vehicle Speed (km/h)')
19 grid on
20 % ----- Graph 2 -----
21 subplot(2,1,2)
22 plot(speed,range,'-s','LineWidth',2,'Color',"g")
23 title('192512146 Vehicle Speed vs Communication Range')
24 xlabel('Vehicle Speed (km/h)')
25 ylabel('Communication Range (m)')
26 grid on

```



Result:

- 1.The Throughput (Mbps) of the 5G communication system increased with increasing Signal-to-Noise Ratio (SNR), demonstrating improved data transmission performance.
- 2.The simulation showed that higher SNR (dB) reduced transmission errors and enhanced the reliability and quality of 5G communication.